JS Objects

- JavaScript provides several built-in ways to store, organize, and work with data.
- In this topic, we learn the commonly used JavaScript objects and object-like structures: Object, Array, String, Date, Math, Map, and Set.

1. Object

An object is a collection of related data stored as key-value pairs. Each key is called a property, and the value can contain data such as a string, number, Boolean, array, or another object.

Creating an Object

```
let student = {
   name: "ABC",
   age: 20,
   course: "JavaScript"
};
```

Accessing Object Properties

Object properties can be accessed using dot notation or bracket notation.

```
console.log(student.name);
console.log(student["age"]);
```

Output:
ABC
20

Adding and Updating Properties

```
student.city = "Vijayawada";
student.age = 21;
```

```
console.log(student.city);
console.log(student.age);
```

- A new property can be added using the object name and property name.
- An existing property can be updated by assigning a new value.

Deleting an Object Property

```
delete student.city;
```

The delete operator removes a property from an object.

2. Array

An array is an ordered collection of values. It is used when multiple values need to be stored under one variable.

```
let marks = [80, 75, 90, 85];
```

Accessing Array Elements

Array indexing starts from 0.

```
console.log(marks[0]);
console.log(marks[2]);
```

Output:
80
90

Common Array Properties and Methods

Property / Method	Purpose	Example
length	Returns the number of elements.	marks.length
push()	Adds an element at the end.	marks.push(95)
pop()	Removes the last element.	marks.pop()
shift()	Removes the first element.	marks.shift()
unshift()	Adds an element at the beginning.	marks.unshift(70)

3. String

A string represents a sequence of characters or text. JavaScript provides several properties and methods for working with strings.

let message = "Hello JavaScript";

Accessing String Characters

console.log(message[0]);

Output: H

Common String Properties and Methods

Property / Method	Purpose	Example
length	Returns the number of characters.	message.length
toUpperCase()	Converts text to uppercase.	message.toUpperCase()
toLowerCase()	Converts text to lowercase.	message.toLowerCase()
charAt()	Returns the character at a position.	message.charAt(0)
includes()	Checks whether text is present.	message.includes("JavaScript")

4. Date

The Date object is used to work with dates and times in JavaScript.

let currentDate = new Date();
console.log(currentDate);

new Date() creates a Date object representing the current date and time.

Creating a Specific Date

```
let date = new Date("2026-10-03");
console.log(date);
```

Common Date Methods

Method	Purpose	Example
getFullYear()	Returns the year.	date.getFullYear()
getMonth()	Returns the month index.	date.getMonth()
getDate()	Returns the day of the month.	date.getDate()
getDay()	Returns the day of the week index.	date.getDay()
getHours()	Returns the hour.	date.getHours()
getMinutes()	Returns the minutes.	date.getMinutes()

Note: getMonth() returns a zero-based month index, where January is 0 and December is 11.

5. Math

Math is a built-in JavaScript object that provides mathematical constants and methods. It is used for common mathematical calculations.

```
console.log(Math.PI);
console.log(Math.sqrt(25));
console.log(Math.pow(2, 3));
```

Output:

```
3.141592653589793
5
8
```

Common Math Properties and Methods

Property / Method	Purpose	Example
Math.PI	Returns the value of π.	Math.PI

Math.sqrt()	Returns the square root.	Math.sqrt(25)
Math.pow()	Returns a number raised to a power.	Math.pow(2, 3)
Math.round()	Rounds to the nearest integer.	Math.round(4.6)
Math.floor()	Rounds down.	Math.floor(4.9)
Math.ceil()	Rounds up.	Math.ceil(4.1)
Math.max()	Returns the largest value.	Math.max(10, 20, 30)
Math.min()	Returns the smallest value.	Math.min(10, 20, 30)
Math.random()	Returns a random number from 0 up to, but not including, 1.	Math.random()

6. Map

A Map is a collection of key-value pairs. Unlike a regular object, a Map can use different types of values as keys.

```
let student = new Map();

student.set("name", "ABC");
student.set("age", 20);

console.log(student.get("name"));
console.log(student.get("age"));
```

Output:
ABC
20

Common Map Methods

Method	Purpose	Example
set()	Adds or updates a key-value pair.	student.set("age", 21)
get()	Returns the value for a key.	student.get("age")
has()	Checks whether a key exists.	student.has("name")

delete()	Removes a key-value pair.	student.delete("age")
clear()	Removes all key-value pairs.	student.clear()
size	Returns the number of entries.	student.size

7. Set

A Set is a collection of unique values. Duplicate values are automatically ignored.

```
let numbers = new Set();

numbers.add(10);
numbers.add(20);
numbers.add(10);

console.log(numbers);
```

The value 10 is stored only once because a Set does not allow duplicate values.

Common Set Methods

Method	Purpose	Example
add()	Adds a value.	numbers.add(30)
has()	Checks whether a value exists.	numbers.has(20)
delete()	Removes a value.	numbers.delete(20)
clear()	Removes all values.	numbers.clear()
size	Returns the number of unique values.	numbers.size